Violet Xiang

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Education

Stanford University Expected Jan 2027
Ph.D. in Psychology
Research focus: LLM post-training, reinforcement learning, reward modeling, and computational cognitive science.

Indiana University 2018–2020
M.S. in Computer Science

Indiana University 2011–2016
B.S. in Informatics, B.A. in Mathematics

Industry Research

Radical Numerics Oct 2025–Jan 2026
Member of Technical Staff Intern
- Built an end-to-end alignment and benchmarking pipeline for genomic foundation models, covering preference pair curation, distributed DPO training orchestration, and downstream generative DNA-sequence evaluation.

Toyota Research Institute Jun-Sep 2025
Research Scientist Intern, Human-AI Interactive Learning
- Formulated a likelihood-gain objective for training corrective teacher models, scoring feedback by its marginal increase in a frozen student model's likelihood on reference solutions.
- Evaluated sparse final-answer rewards against dense intermediate feedback, characterizing failure modes and error propagation during transfer to long reasoning traces.

Publications

*Equal contribution.

LLM Post-Training, Reward Modeling, and Evaluation

V Xiang, M Khalifa, I Wu, Y Shen, J Leskovec, N Haber, A Kumar. Test-Time Self Improvement by Scaling Unsupervised World Internalization. *Under review at ICLR 2027.*
- Introduced U-WIN, an unsupervised self-improvement framework where agents explore unseen environments, synthesize structured notes, and self-train on grounded trajectories without task labels or external rewards.
- Demonstrated near log-linear scaling from 10M to 100M training tokens across legal, code, and formal math benchmarks (Harvey, StudyBench, Equational Theories) using Qwen 9B and 27B models.
- Increased environment knowledge recall from 20.6% to 99.8%; demonstrated that self-authored notes retain state significantly better than token-matched raw observations.

V Xiang, A Setlur, C Blagden, N Haber, A Kumar. ExpRL: Exploratory RL for LLM Mid-Training. *COLM 2026* | *code*.
- Developed ExpRL, an online RL mid-training method using reference solutions to scaffold LLM judges with dense process- and outcome-level rewards.
- Outperformed baseline models on held-out AIME 2026 pass@1 (Qwen3-4B) by +3.6 points after ExpRL priming and +4.7 points after downstream sparse-GRPO post-training.

D Fein*, M Lamparth*, **V Xiang**, M J Kochenderfer, N Haber. One Bias After Another: Mechanistic Reward Shaping and Persistent Biases in Language Reward Models. *ICML 2026* | *code*.
- Dissected five persistent biases across leading Bradley-Terry (BT) reward models: length, uncertainty, position, sycophancy, and style.
- Quantified severe vulnerabilities in leading models: a 22.6-point drop on hedged correct answers, up to 60% agreement with false user premises, and positional variance up to 28 points.
- Mitigated length, uncertainty, and position biases via null-space probe projections, eliminating bias without retraining or degradation on RewardBench-2.

D Fein*, S Russo*, **V Xiang***, K Jolly, R Rafailov, N Haber. LitBench: A Benchmark and Dataset for Reliable Evaluation of Creative Writing. *EACL 2026, Long Paper* | *dataset*.
- Constructed LitBench, a creative-writing evaluation benchmark featuring a 43k-pair training set and an expert-annotated, debiased 2.5k-pair test set.
- Trained BT and generative reward models that surpassed top zero-shot LLM judges by 5.0 points in human preference alignment.
- Validated RM calibrations via an online human evaluation study on generated narratives.

V Xiang, C Blagden, R Rafailov, N Lile, S Truong, C Finn, N Haber. Just Enough Thinking: Efficient Reasoning with Adaptive Length Penalties Reinforcement Learning. *NeurIPS 2025 Efficient Reasoning Workshop Spotlight.*
- Formulated an adaptive length penalty for GRPO (evaluated on DeepScaleR-1.5B), halving token length at matched accuracy across MATH-500, AIME, and OlympiadBench while allocating 5.35× more compute to difficult prompts.

T Hua, H Hua, **V Xiang**, B Klieger, S Truong, W Liang, F-Y Sun, N Haber. ResearchCodeBench: Benchmarking LLMs on Implementing Novel Machine Learning Research Code. *NeurIPS 2025 Datasets & Benchmarks Track, Spotlight* | *website*.
- Engineered an open-source execution harness across 212 challenge tasks from 20 ML papers; evaluated 32 frontier models, with the strongest achieving only 33.0% on the HARD split and 43 tasks remaining unsolved across all models.

V Xiang, C Snell, K Gandhi, A Albalak, A Singh, C Blagden, D Phung, R Rafailov, N Lile, D Mahan, L Castricato, J-P Franken, N Haber, C Finn. Towards System 2 Reasoning in LLMs: Learning How to Think With Meta Chain-of-Thought. *Preprint, 2025.*

- Formalized a training framework for System 2 reasoning combining linearized search traces, process supervision, and RL; established empirical evidence of spontaneous in-context search in frontier reasoning models.

Multi-Agent Learning

L Cross, **V Xiang**, A Bhatia, D L K Yamins, N Haber. Hypothetical Minds: Scaffolding Theory of Mind for Multi-Agent Tasks with Large Language Models. *ICLR 2025, Poster.*

- Designed and implemented an LLM Theory-of-Mind agent for multi-agent cooperation, outperforming standard MARL baselines on 3 of 4 environments and 6 of 8 eight-agent scenarios across 30 Melting Pot tasks.

V Xiang, L Cross, J-P Fränken, N Haber. From Centralized to Self-Supervised: Pursuing Realistic Multi-Agent Reinforcement Learning. *NeurIPS 2023 ALOE Workshop, Poster.*

- Benchmarked centralized, decentralized, asynchronous, and self-supervised MARL pipelines, demonstrating that centralized training assumptions systematically overestimate performance in realistic mixed-motive settings.

Human and Machine Learning

B Long*, R Sparks*, **V Xiang***, S Stojanov*, et al.. The BabyView Dataset: High-Resolution Egocentric Videos of Infants' and Young Children's Everyday Experiences. *Computational Cognitive Neuroscience 2025 Proceedings.*

- Curated an 868-hour developmental egocentric-video dataset (ages 5mo–3yr) with expert annotations and evaluation suites for transcription, diarization, pose estimation, and out-of-distribution transfer.

C Zhuang*, **V Xiang***, Y Bai, X Jia, N Turk-Browne, K Norman, J J DiCarlo, D L K Yamins. How Well Do Unsupervised Learning Algorithms Model Human Real-Time and Lifelong Learning?. *NeurIPS 2022 Datasets & Benchmarks Track.*

- Benchmarked self-supervised visual learning against infant visual streams, identifying continual memory mechanisms as essential to prevent representation collapse on sparse, low-diversity inputs.

L Yuan, **V Xiang**, D Crandall, L Smith. Learning the Generative Principles of a Symbol System from Limited Examples. *Cognition, 2020.*

Privacy and Security

Y Liu, **V Xiang**, E J Seong, A Kapadia, D S Williamson. Defending Against Microphone-Based Attacks with Personalized Noise. *Proceedings on Privacy Enhancing Technologies, 2021.*

Technical Skills

Library & Tools: FSDP, Docker, vLLM, SGLang, Ray/RLlib, SLURM

Deep Learning Frameworks: PyTorch, Tensorflow, Keras

Programming Languages: Python, C++, C, JavaScript

Teaching

2023-25	Teaching Assistant, Neural Network Models of Cognition, Stanford University
2023	Teaching Assistant, Introduction to Statistical Methods, Stanford University
2021	Teaching Assistant, Longitudinal Design and Data Analysis, Stanford University
2019	Mentor, Proactive Health Informatics REU, Indiana University
2019	Teaching Assistant, Computer Vision, Indiana University
2018	Teaching Assistant, Introduction to Data Structures and Algorithms, Indiana University